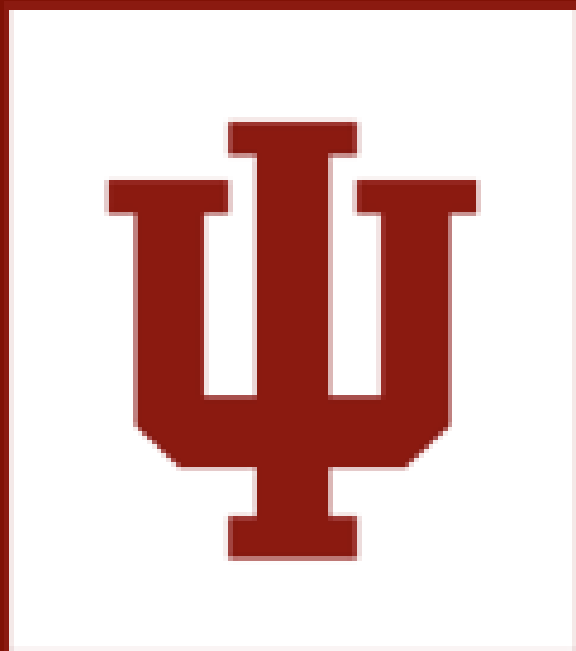




Egocentric Hand Detection in Infant Video

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Abstract

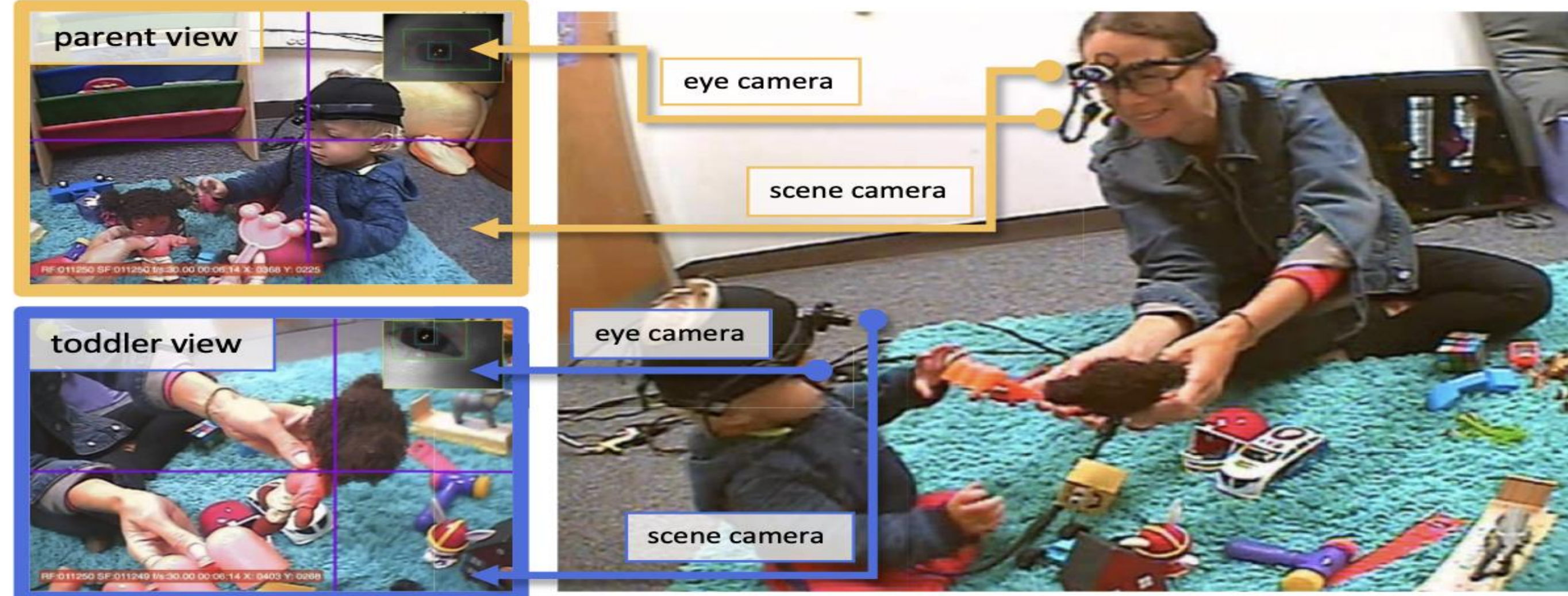


Fig 1. Egocentric camera views showing infant perspective during hand detection [1].

Understanding how infants learn requires knowing what they see and touch from their own perspective. Traditional developmental psychology studies observe infants from an external camera, missing the infant's first-person view entirely. Egocentric (head-mounted) cameras capture exactly this, but detecting infant hands in such footage is an unresolved computer vision problem. Perspective shifts constantly, hands appear at unpredictable scales, and lighting and backgrounds are uncontrolled. This project builds an end-to-end pipeline for egocentric infant hand detection and object interaction analysis.

Dataset & Annotation

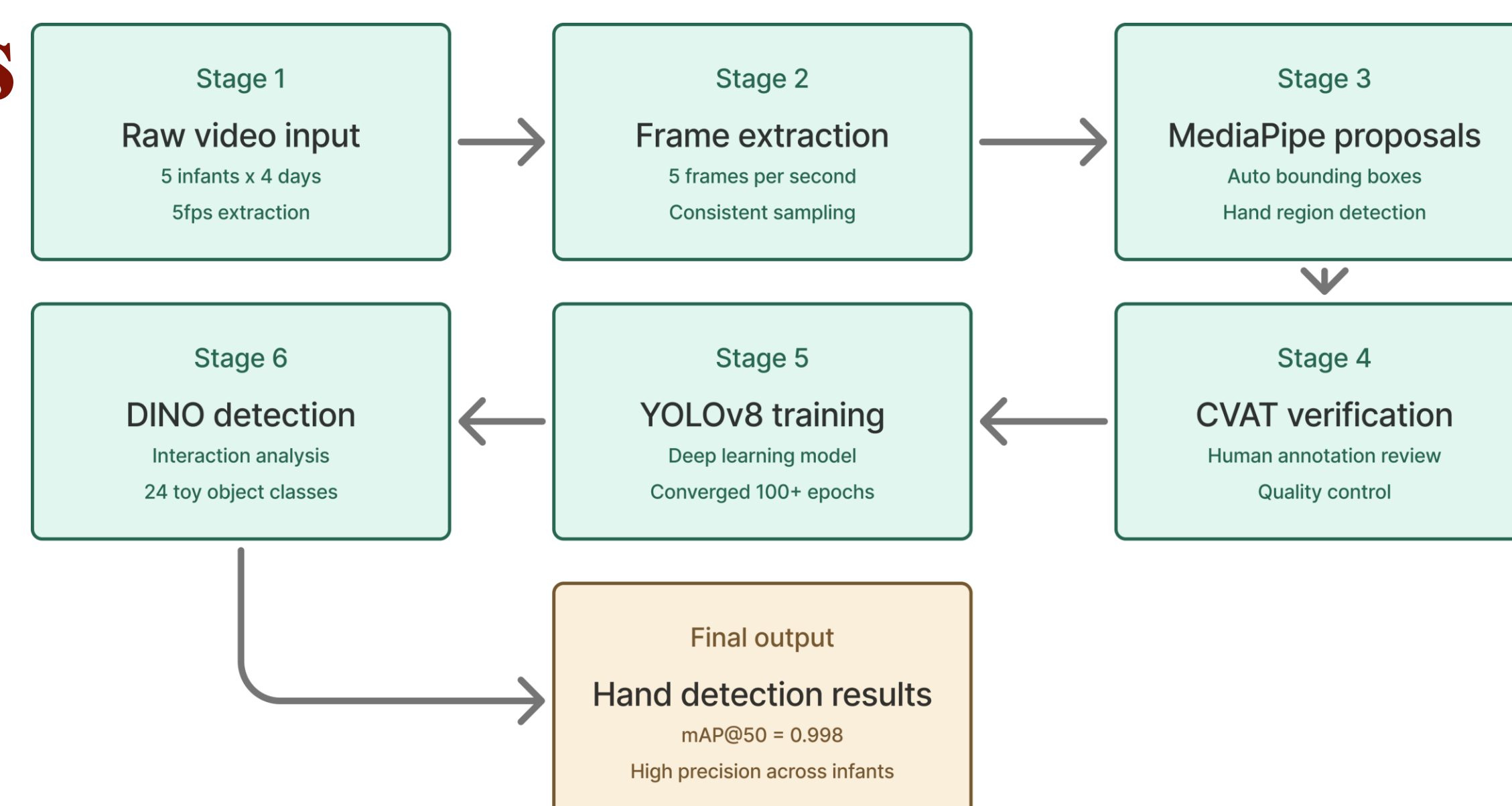
Hand detection dataset:

- Dataset consists of 5 infants, each recorded over 4 days
- Video frames were extracted at 5 frames per second (5 FPS)
- Initial bounding boxes were generated using MediaPipe
- All frames were manually verified and corrected using CVAT

Object detection dataset:

- 30 frames per object were extracted from the videos
- These frames were used to generate embeddings using DINOv2
- The embeddings were used to build class prototypes / reference representations for each object

Methods & Pipeline



Hand Detection YOLOv8 Training



Fig 1. YOLOv8 hand detection on Day 4 test footage. Bounding boxes with confidence scores show simultaneous detection of two hands in a single egocentric frame.

Object Detection Results

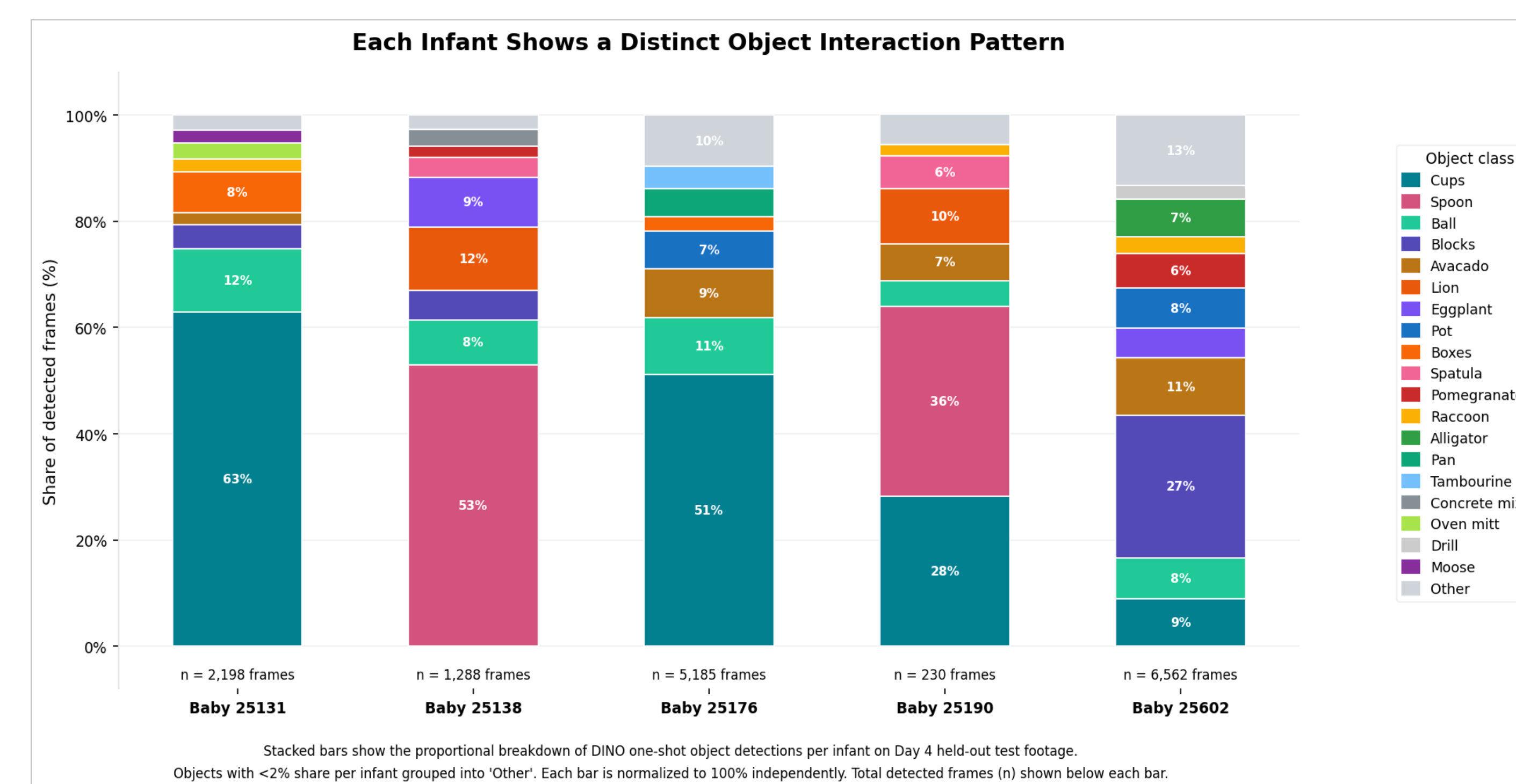
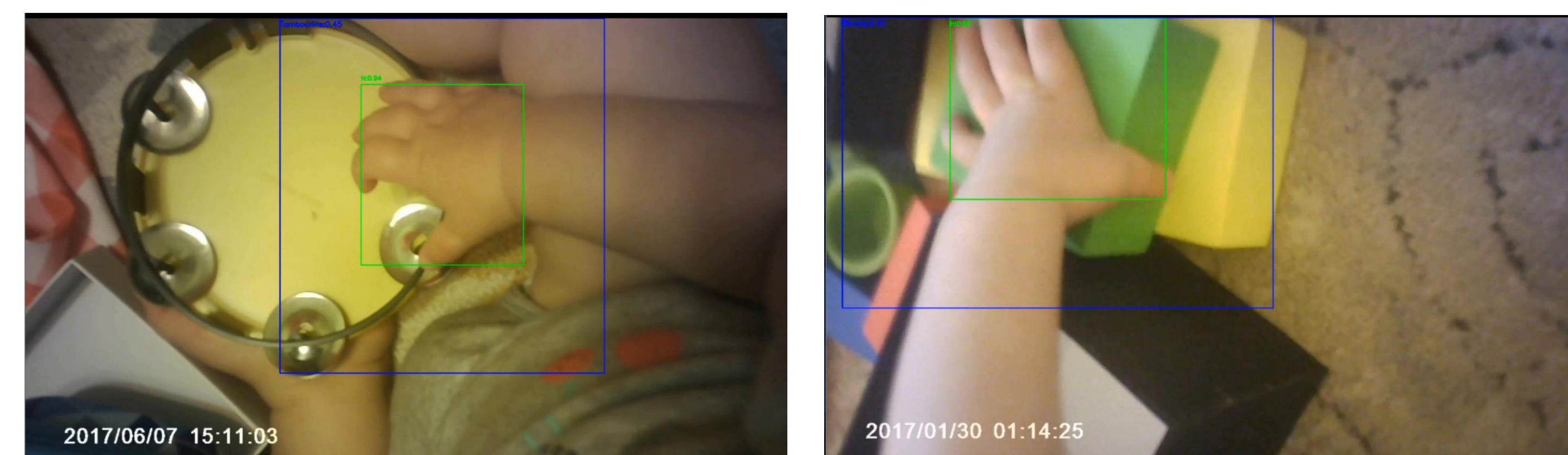


Fig 2. Stacked bars show the proportional breakdown of DINO one-shot object detections per infant on Day 4 held-out test footage..

Key Results

Hand detection :

- YOLOv8 training converged stably over 100 epochs with no overfitting validation losses closely tracked training.
- Final metrics: Precision ≈ 0.99 , Recall ≈ 0.99 , mAP@50 ≈ 0.998 , mAP@50-95 ≈ 0.89 .
- The model generalized successfully to footage from new environments and infants not seen during training.

Object detection :

- The pipeline detected interactions across all 5 infants with 24 toy object classes on Day 4 test footage.
- Interaction patterns varied across infants — Baby 25176 showed heavy cup and ball interaction, Baby 25602 engaged primarily with blocks, and Baby 25138 showed high spoon interaction.
- These differences reflect genuine behavioral variation worth studying developmentally.

Limitations

- DINO one-shot detection correctly identifies that a target object is present in a frame but tends to localize detections around the hand region rather than the object itself.
- This is a known limitation of one-shot detection in cluttered, low-resolution egocentric footage.
- Results should be interpreted as hand-region object associations rather than precise object localization.

Conclusions

- We built and validated an end-to-end egocentric infant hand detection pipeline, with strong generalization across recording days and new environments.
- Object interaction analysis reveals meaningful behavioral differences across infants.
- The primary remaining challenge is precise object localization within detected hand regions.

Reference

- [1] Bambach, S., Crandall, D.J., Smith, L.B., Yu, C. (2018). Toddler-Inspired Visual Object Learning. Advances in Neural Information Processing Systems 31 (NeurIPS 2018).